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TOPIC Calculate Covariance & Correlation DATE

r = Correlation = $\frac{Cov(P, Q)}{\sigma_p \cdot \sigma_q}$

P	Q
1	2
2	4
3	5
4	4
5	5

← Two features

$(P-\bar{P})$	$(P-\bar{P})^2$	$(Q-\bar{Q})$	$(Q-\bar{Q})^2$
-2	4	-2	4
-1	1	0	0
0	0	1	1
1	1	0	0
2	4	1	1

$$Cov(P, Q) = \frac{1}{n} \sum_{i=1}^{n} \left((P^{(i)} - \bar{P})(Q^{(i)} - \bar{Q}) \right)$$

$$Cov(P, Q) = \frac{1}{5} \sum_{i=1}^{5} (4 + 0 + 0 + 0 + 2) = \frac{6}{5}$$

$$\sigma_p = \sqrt{\frac{1}{5} \sum_{i=1}^{n} (P^i - \bar{P})^2} = \sqrt{\frac{1}{5} 10} = \sqrt{2}$$

$$\sigma_q = \sqrt{\frac{1}{5} \sum_{i=1}^{5} (Q - \bar{Q})^2} = \sqrt{\frac{1}{5} \times 6} = \frac{6}{5}$$

$$r = \text{Correlation} = \frac{\frac{6}{5}}{\sqrt{\frac{6}{5}} \times \sqrt{2}} = \frac{\sqrt{\frac{6}{5}} \times \cancel{\sqrt{\frac{6}{5}}}}{\cancel{\sqrt{\frac{6}{5}}} \times \sqrt{2}} = \frac{\sqrt{\frac{6}{5}}}{\sqrt{2}} = 0.774$$

$$Cov(P, Q) = \frac{6}{5}$$

correlation = 0.774

Compute the Eigen Values and Eigen vector

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

① characteristic Equation

$$\det(A - \lambda I) = 0$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$(2-\lambda)^2 - 1^2 = 0$$

$$(2-\lambda)^2 = 1$$

$$2 - \lambda = \pm 1$$

$$\boxed{\begin{matrix} \lambda = 3 \\ \lambda = 1 \end{matrix}}$$ Eigen values

Now substituting

For $\lambda = 3$

$$A - 3I = \begin{bmatrix} 2-3 & 1 \\ 1 & 2-3 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$-x + y = 0$$

$$x = y$$

If $x = 1$

$$\vec{V}_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\lambda = 1$

$$A - I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$x + y = 0 \Rightarrow y = -x$$

$$\vec{V}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

eigen values :

$\lambda_1 = 3 \qquad \lambda_2 = 1$

$$\vec{V}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \vec{V}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$